



TrainingIT

A Web-Based Application for Selling Programming Courses

Raul Veca

Supervisor: Lect. Dr. Sebastian Ștefănița

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Problem and Motivation

One storefront and one back office - normally two different systems

The customer side. A learner expects what any modern shop offers: a clear catalogue, prices and levels, reviews they can trust, a purchase in one click, and a private tutoring session booked at a time that suits them.

The business side. Behind the storefront the company must know who its leads are and how close each is to buying, keep a reliable purchase history, issue correct invoices with negotiated discounts, read the health of the business, and handle corporate clients who send whole teams.

The integration gap. These two sides are normally separate tools that must be synchronised. Every synchronisation point is a place where data can drift, and manual reconciliation is slow and error-prone.

Problem statement

Build a single web application in which the customer-facing storefront and the business-facing CRM are not two integrated systems but one system - so that every customer action is captured exactly once and automatically drives enrolment, invoicing, lead scoring, auditing and support.

It must also stay approachable for non-technical staff, keep the two roles strictly separated, and remain fully functional when optional services are switched off.

Related Work Gap

Existing tools solve parts of the problem, not the whole workflow

Each family of tools covers one side well; none unifies a course storefront and a CRM in a single product.

Tool family	Strength	Limitation
Course marketplaces e.g. Udemy, Coursera	Excellent storefront, catalogue, reviews and checkout	Aggregators: the provider owns no customer relationship and has no integrated CRM
Learning management e.g. Moodle	Strong at delivering and tracking course content	Built around content, not selling; commerce and CRM are plug-ins bolted on
CRM platforms e.g. Salesforce, HubSpot	Powerful contacts, pipelines, lead scoring and reporting	Generic: no course catalogue, trainer calendar or public storefront
E-commerce engines e.g. WooCommerce	Self-hosted and able to sell anything, including courses	A course is just a product: no lead stages, tutoring sessions or corporate training

TrainingIT unifies both sides in one application built on one shared data model, and specialises the generic CRM notions for the training domain: course sessions, trainer calendars, per-hour tutoring with employee discounts, and course-level click-through analytics.

Applied contribution

TrainingIT is a functional web application: a course marketplace whose every customer action is recorded exactly once in a complete CRM back office, on a Java domain layer organised around classic design patterns and an event-driven core, with an optional Claude-powered AI layer.

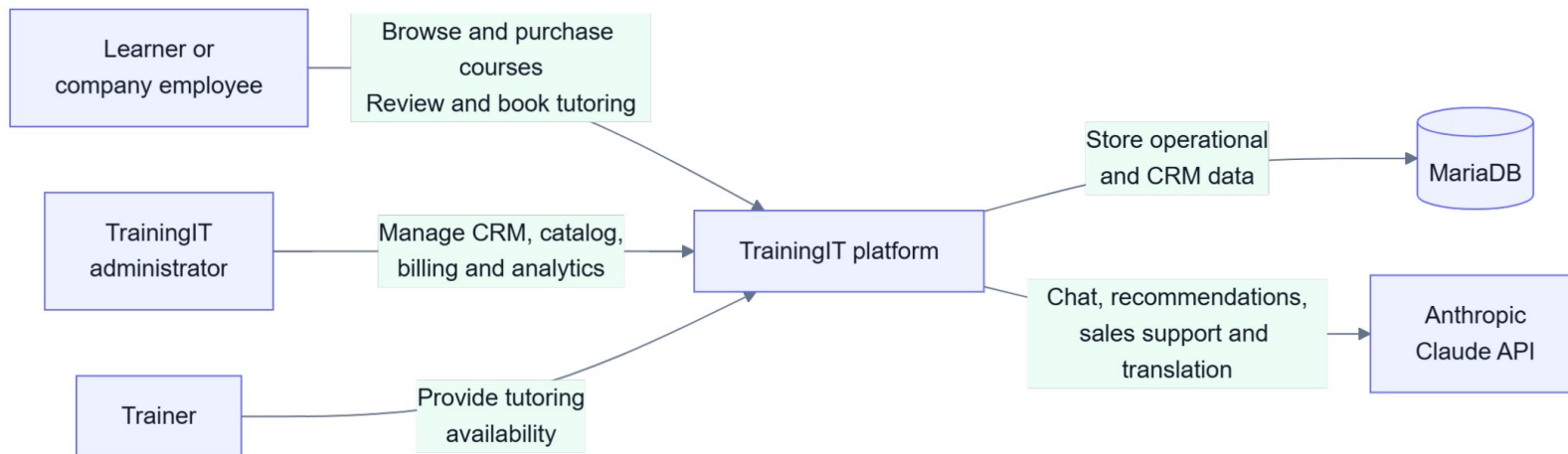
Objectives

- A public client portal: browse, purchase, review, book one-to-one tutoring.
- An administrator portal with seven management tabs.
- A domain layer on design patterns and an event-driven core, so one action propagates into several automatic reactions.
- An optional AI assistant that degrades gracefully.
- All operational and CRM data persisted in MariaDB.

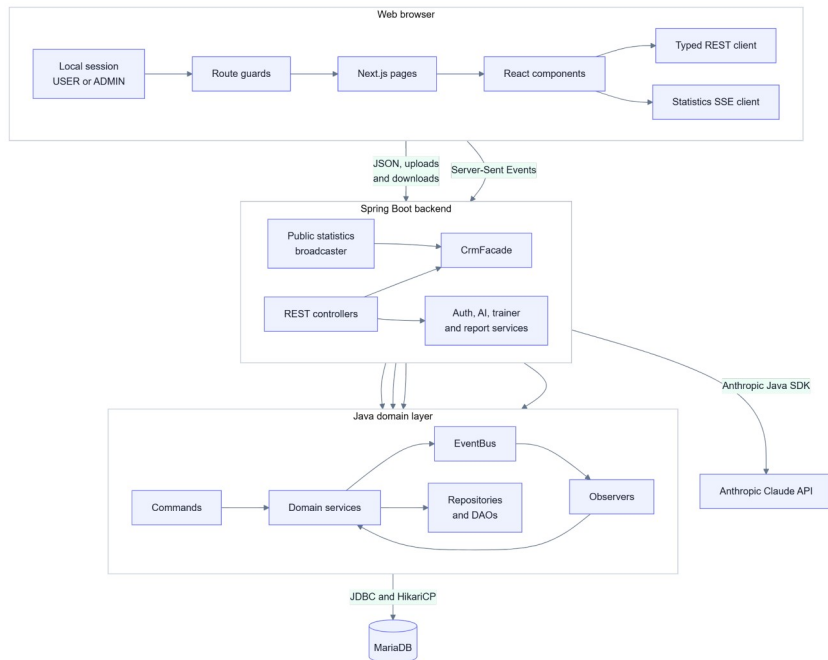
Design boundary

The goal is not a general-purpose e-commerce platform, nor a general-purpose CRM.

The goal is one coherent product for a training company: a storefront a learner can use without instructions, over a CRM core a small team can actually run - where the AI layer is strictly additive and never a dependency.



Learners and partner-company employees buy and review courses; administrators run the CRM, catalogue, billing and analytics; trainers provide tutoring availability. MariaDB stores everything; the Anthropic Claude API is the only other dependency - and it is optional.



Presentation - Next.js 16, React, Tailwind v4

Pages, route guards from the local session, a typed REST client and a Server-Sent Events client for live statistics.

Application - Spring Boot 3.5, Java 17

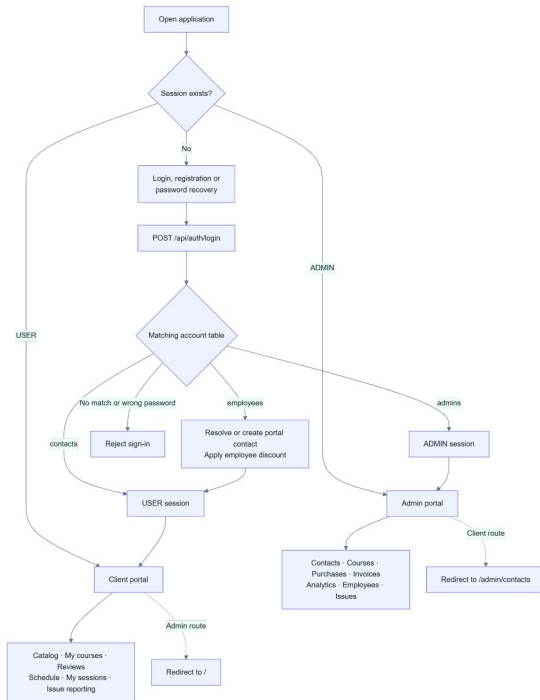
REST controllers under /api, the auth, AI, trainer and report services, and a public statistics broadcaster.

Domain - plain Java

Commands, domain services, an EventBus with observers, and repositories. CrmFacade is the single entry point over all of it.

Persistence - MariaDB via JDBC and HikariCP

Contacts, enrolments, invoices and issues survive restarts; the schema is created and updated at start-up.



One login, two experiences.

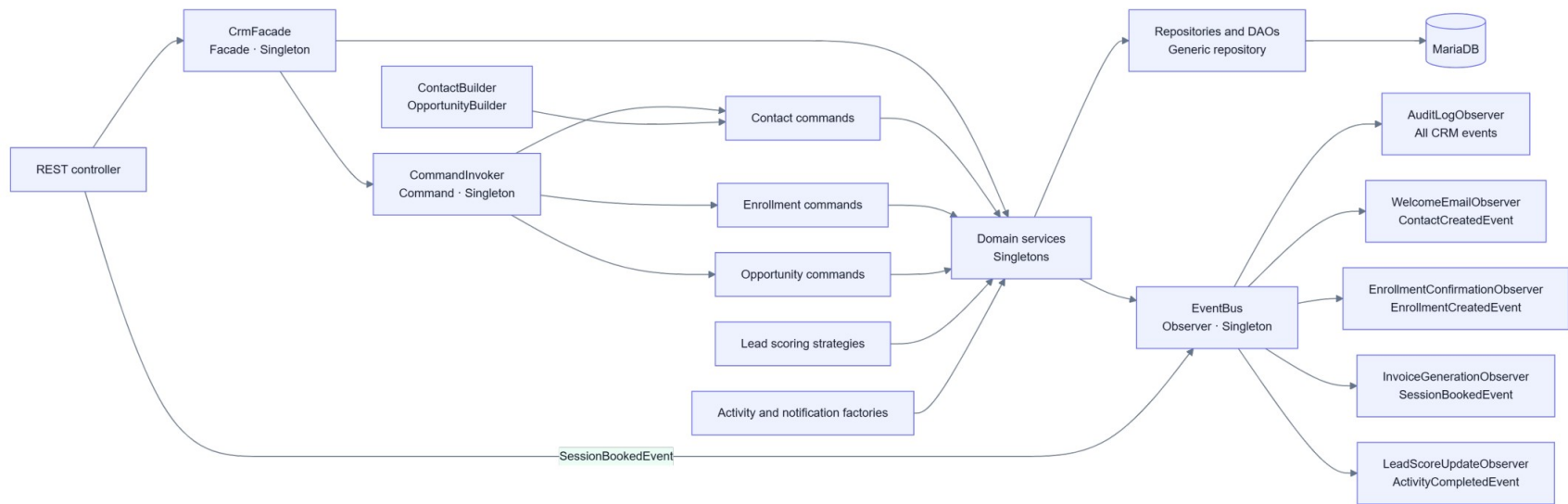
The back end looks the e-mail up across the contact, administrator and employee tables and verifies the password. The role it returns decides the destination: a USER lands on the public portal, an ADMIN on /admin.

Enforced separation

If an administrator requests a client route, or a client requests an administrator route, the application redirects them back to their own portal. An unauthenticated visitor reaches only the login, registration and password-recovery screens.

Employees are resolved on the way in

When a partner-company employee signs in, the system resolves or creates their portal contact and applies the negotiated employee discount automatically - so corporate pricing needs no separate account type.

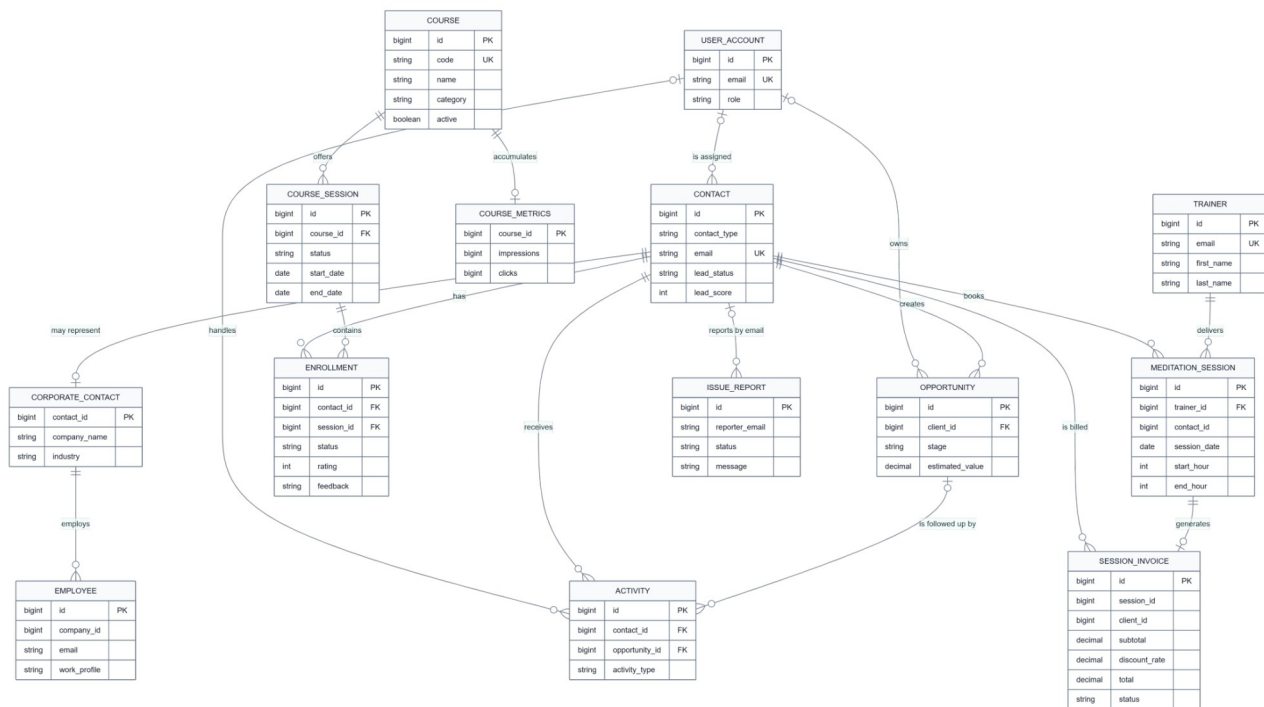


Facade - one entry point over the CRM · Command - every write is an executable command · Observer - an EventBus fans one event out to invoice generation, audit logging, confirmations and lead re-scoring · Strategy - individuals and companies score differently · Factory and Builder - notifications, activities, contacts and opportunities



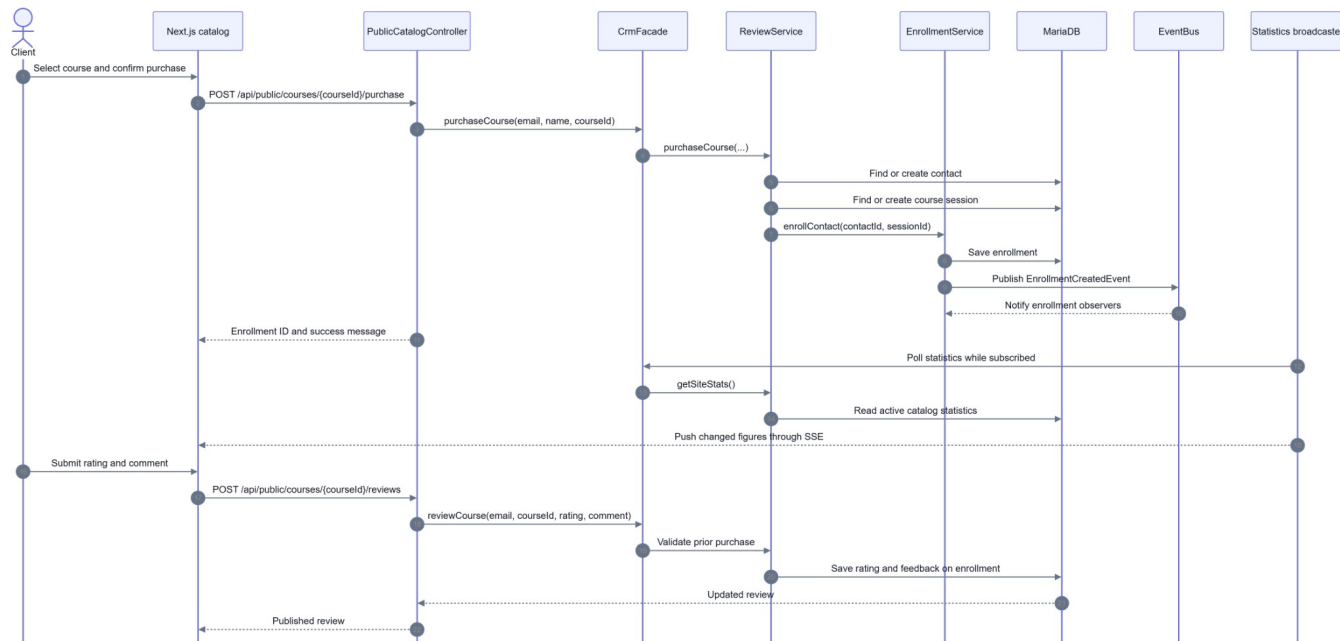
Data Model

One shared schema for the storefront and the CRM



One Click, Several Reactions

Purchase and review, from the client's confirmation to live statistics



A single purchase finds or creates the contact and the course session, saves the enrolment, publishes an event, notifies the observers, and updates the live figures over SSE - with no manual reconciliation anywhere.



The AI Layer

Optional intelligence, powered by the Anthropic Claude API

Conversational assistant `/api/ai/chat`

Answers questions about courses, enrolment and corporate training, using the live catalogue as context.

Recommendation quiz `/api/ai/recommendations/visitor`

Matches a visitor's interests, level and goal to courses, each with a fit score and a reason.

Per-company recommendations `/api/ai/recommendations/company`

Analyses employee roles and interests to propose a training programme for a whole team.

Live page translation `/api/ai/translate`

Translates the interface into any language, preserving brand names, people, e-mails and numbers.

Strictly additive: with no API key the buttons disappear

Find your next course

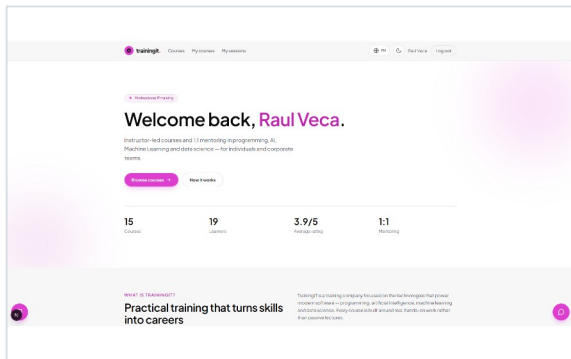
Not sure where to start? Take the quick quiz for a recommendation, or browse by category.

All four features share one Claude client and one API key. I implemented the prompts, the REST endpoints, the structured JSON response handling and the matching interfaces - and made sure the rest of the application stays fully operational when AI is not configured.



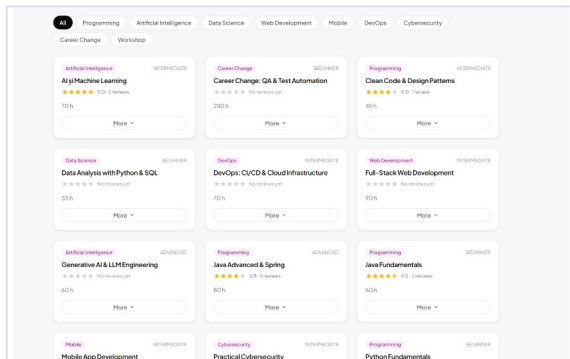
Application Demo

The running application



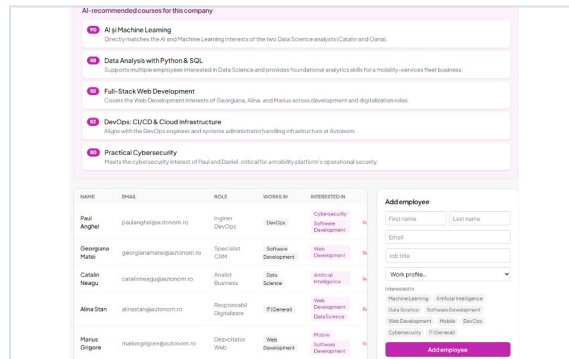
Authenticated learner home

Personalised greeting and live catalogue statistics over SSE



Course catalogue

Category filters; each card counts an impression and a click for CTR



Administrator · Employees

Company teams, Excel import and AI course recommendations

The same session drives both sides: a purchase made on the catalogue appears immediately in the administrator's Purchases tab, a booking generates its invoice by itself, and a reported issue lands straight in the Issues tab.



Testing and Validation

Correctness here is about propagation, not isolated algorithms

Input validation

Validated on two levels. The client checks missing fields, password confirmation and the mandatory GDPR consent. On the server, payloads pass Spring Validation, and contact data additionally goes through a Chain-of-Responsibility validator that checks the e-mail, the phone number, the GDPR flag and the contact type in sequence - so an invalid contact never reaches the database.

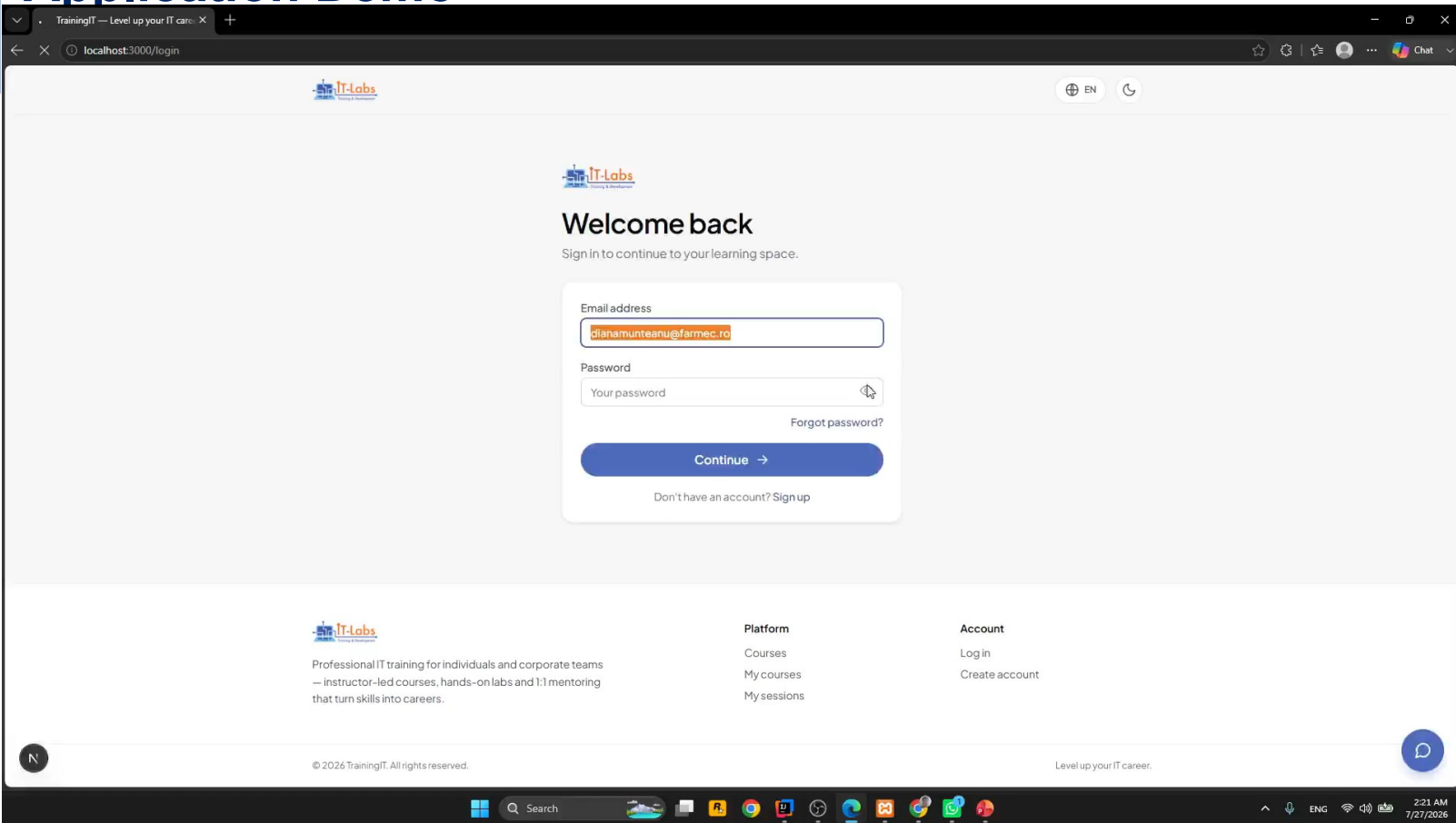
Robustness by design

Invoice generation is idempotent, so a repeated event cannot produce a duplicate. Observer failures are isolated: an error while generating an invoice is logged but does not roll back the booking that triggered it. The AI layer degrades gracefully. Deletions in the Issues tab are guarded by an explicit "this cannot be undone" confirmation.

Functional testing

The main user journeys were tested manually against a running instance backed by a real MariaDB database: registration creating a lead; login routing USER and ADMIN correctly; a one-click enrolment appearing in Purchases; a review updating the average rating live; a booking producing an invoice with the right discount; exports to CSV, Excel and PDF.

The project also includes the Spring Boot test harness as a foundation; extending it into full unit and integration coverage is identified as future work.



The screenshot shows a web browser window displaying the IT-Labs login page. The browser's address bar shows 'localhost:3000/login'. The page features the IT-Labs logo at the top left and a 'Welcome back' message. Below the message, there is a login form with fields for 'Email address' (containing 'oliana.munteanu@farmec.ro') and 'Password' (with a toggle for visibility). A 'Continue' button is positioned below the password field, and a 'Forgot password?' link is to its right. At the bottom of the form, there is a link for users who don't have an account. The footer of the page includes the IT-Labs logo, a description of the platform, and links for 'Platform' (Courses, My courses, My sessions) and 'Account' (Log in, Create account). The bottom of the browser window shows the Windows taskbar with various application icons and the system clock indicating 2:21 AM on 7/27/2026.

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localhost:3000/login

EN

IT-Labs
Training & Development

Welcome back

Sign in to continue to your learning space.

Email address

oliana.munteanu@farmec.ro

Password

Your password

Forgot password?

Continue →

Don't have an account? Sign up

IT-Labs
Training & Development

Professional IT training for individuals and corporate teams
— Instructor-led courses, hands-on labs and 1:1 mentoring
that turn skills into careers.

Platform

- Courses
- My courses
- My sessions

Account

- Log in
- Create account

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Level up your IT career.

2:21 AM
7/27/2026



Conclusion and Future Work

One product where the storefront and the CRM are the same system

Implemented

- A client portal: catalogue, one-click purchase, star reviews, 1:1 booking with calendar and payment, issue reporting.
- A seven-tab administrator portal: contacts, courses, purchases, invoices, analytics, employees, issues.
- An event-driven Java domain layer on classic design patterns, with automatic invoicing and lead scoring.
- Live statistics over SSE, Excel import, and CSV / Excel / PDF export.
- An optional Claude layer: assistant, recommendation quiz, per-company advice, live translation.

Limitations and next steps

- Automated test suite - unit and integration tests over the commands, observers, scoring strategies and REST endpoints, enabling continuous integration.
- Stronger authentication - token-based sessions, password hashing at rest, e-mail verification and a genuine reset flow with expiring links.
- Richer analytics - historical trends, cohort analysis and exportable dashboards beyond point-in-time metrics.
- Deeper AI - conversation memory and AI summaries of support issues.
- Deployment and scaling - containerising the two services for a cloud deployment.

A visitor becomes a tracked lead on registration and a customer on the first purchase, and every action after that propagates into the administration portal by itself. That is the result: one click, several reactions, no reconciliation.